PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

19 Oct – 25 Oct 2026 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
19 Oct T2026-10-19	Maths Weekly test	___ / 40	Retest: 22 Oct
23 Oct D2026-10-23	Daily mixed Daily test	___ / 10	Retest: 26 Oct
24 Oct D2026-10-24	Daily mixed Daily test	___ / 10	Retest: 27 Oct
25 Oct D2026-10-25	Daily mixed Daily test	___ / 10	Retest: 28 Oct

Maths · Weekly Test 1 · Answer key

Monday, 19 October 2026 · 40 marks

Test ID: T2026-10-19

Class 9 · Ch 1 Coordinates · Ch 2 Linear Polynomials · Ch 5 Circles (5.1–5.8)

Q1 [1]

Question: A point on the y-axis, 5 units below the origin, has coordinates:

(a) (5, 0) (b) (–5, 0)

(c) (0, –5) (d) (0, 5)

Expected answer / marking points: (c) (0, –5)

Q2 [1]

Question: The coordinates of every point on the x-axis are of the form:

(a) (0, y) (b) (x, 0)

(c) (x, x) (d) (x, y) with x = y

Expected answer / marking points: (b) (x, 0)

Q3 [1]

Question: Which of the following is a linear polynomial?

(a) $x^2 + 1$ (b) 5 – 4y

(c) $x^3 - 2x$ (d) 7

Expected answer / marking points: (b) 5 – 4y (degree 1)

Q4 [1]

Question: The lines y = 2x + 3 and y = 2x – 5 are:

(a) perpendicular (b) parallel

(c) the same line (d) intersecting at (0, 3)

Expected answer / marking points: (b) parallel — same slope 2, different y-intercepts.

Q5 [1]

Question: Linear decay is represented by a straight line with:

(a) positive slope (b) zero slope

(c) negative slope (d) no y-intercept

Expected answer / marking points: (c) negative slope

Q6 [1]

Question: An arc subtends an angle of 70° at the centre of a circle. The angle it subtends at a point on the remaining part of the circle is:

(a) 140° (b) 70°

(c) 35° (d) 110°

Expected answer / marking points: (c) 35° — half of the angle at the centre.

Q7 [1]

Question: ABCD is a cyclic quadrilateral with ∠A = 75°. Then ∠C =

(a) 75° (b) 105°

(c) 115° (d) 15°

Expected answer / marking points: (b) 105° — opposite angles of a cyclic quadrilateral add up to 180°.

Q8 **[1]** **Question:** The angle subtended by a diameter at any point on the circle is:

(a) 45° (b) 60°

(c) 90° (d) 180°

Expected answer / marking points: (c) 90°

Q9 **[1]** **Question:** How many circles can pass through three points that are not on one straight line?

(a) None (b) Exactly one

(c) Exactly two (d) Infinitely many

Expected answer / marking points: (b) Exactly one (the circumcircle).

Q10 **[1]** **Question: A:** Equal chords of a circle are at equal distances from the centre.
R: Of two unequal chords of a circle, the longer chord is closer to the centre.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (b) Both true, but R does not explain A.

Q11 **[1]** **Question: A:** The point (4, –4) lies on the line $y = -x$.
R: For every point on the line $y = -x$, the x-coordinate and the y-coordinate are equal.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (c) A true ($–4 = –4$); R false — on $y = –x$ the coordinates are negatives of each other.

Q12 **[2]** **Question:** You have ₹800 and save ₹250 every month. Write the amount after n months as a linear pattern and find the amount after (i) 6 months (ii) 2 years.

Expected answer / marking points: $A = 800 + 250n$ [1]. (i) $n = 6$: ₹2,300 [0.5]; (ii) $n = 24$: ₹6,800 [0.5].

Q13 **[2]** **Question:** Find the distance of the point P(6, 8) from the origin.

Expected answer / marking points: $OP = \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ units (Pythagoras on the right triangle formed with the axes) [1+1].

Q14 **[2]** **Question:** In a circle of radius 13 cm, a chord is 5 cm away from the centre. Find the length of the chord.

Expected answer / marking points: Half-chord $= \sqrt{(13^2 – 5^2)} = \sqrt{144} = 12$ cm [1]; chord = 24 cm (perpendicular from centre bisects the chord) [1].

Q15 **[2]** **Question:** PQRS is a cyclic quadrilateral with $\angle P = (2x + 10)°$ and $\angle R = (3x – 20)°$. Find x, $\angle P$ and $\angle R$.

Expected answer / marking points: $\angle P + \angle R = 180° \Rightarrow 5x – 10 = 180 \Rightarrow x = 38$ [1]; $\angle P = 86°$, $\angle R = 94°$ [1].

Q16 [3] **Question:** A positive number is 5 times another number. If 21 is added to both numbers, one of the new numbers becomes twice the other new number. Find the numbers.

Expected answer / marking points: Let numbers be x and 5x [0.5]. $5x + 21 = 2(x + 21)$ [1] $\Rightarrow 3x = 21 \Rightarrow x = 7$ [1]. Numbers: 7 and 35 [0.5]. (Check: $56 = 2 \times 28$.)

Q17 [3] **Question:** M(–7, 1) is the midpoint of the segment joining A(3, –4) and B(x, y). Find the coordinates of B.

Expected answer / marking points: $(3 + x)/2 = -7 \Rightarrow x = -17$ [1.5]; $(-4 + y)/2 = 1 \Rightarrow y = 6$ [1.5]. B(–17, 6).

Q18 [3] **Question:** Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle. The distance between them is 7 cm. Find the radius of the circle.

Expected answer / marking points: Let d = distance of the 24 cm chord from centre; the 10 cm chord is at d + 7 [0.5]. $r^2 = 12^2 + d^2 = 5^2 + (d + 7)^2$ [1] $\Rightarrow 144 = 74 + 14d \Rightarrow d = 5$ [1] $\Rightarrow r^2 = 169$, r = 13 cm [0.5].

Q19 [3] **Question:** For the line $y = -3x + 4$: (i) write its slope and y-intercept, (ii) find the point where it cuts the x-axis, (iii) is this linear growth or linear decay?

Expected answer / marking points: (i) slope –3, y-intercept 4 [1]; (ii) $0 = -3x + 4 \Rightarrow (4/3, 0)$ [1]; (iii) linear decay — slope negative [1].

Q20 [4] **Question:**

> A computer screen is 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 pixels has its centre at A(100, 150). Another circular icon of radius 100 pixels has its centre at B(250, 230).

(i) Does any part of the circle with centre A go outside the screen? Justify. [1]
(ii) Does any part of the circle with centre B go outside the screen? Justify. [1]
(iii) Find the distance AB and decide whether the two circles intersect. [2]

Expected answer / marking points: (i) Leftmost x = 100 – 80 = 20, lowest y = 150 – 80 = 70; right 180, top 230 — all within 0–800 and 0–600, so fully inside [1].
(ii) Left 150, bottom 130, right 350, top 330 — fully inside [1].
(iii) $AB = \sqrt{(150^2 + 80^2)} = \sqrt{28900} = 170$ px [1]. Sum of radii = 180 > 170, so the circles intersect [1].

Q21 [5] **Question:** Prove that the angle subtended by an arc at the centre of a circle is twice the angle subtended by it at any point on the remaining part of the circle. Draw a neat figure.

ATHVA / OR

A growing pattern of hexagons is made with matchsticks; each new hexagon shares one side with the previous one (Stage 1 = 1 hexagon, Stage 2 = 2 joined hexagons, ...). (i) Make a table of matchsticks for stages 1 to 4. (ii) Find the rule for stage n. (iii) How many matchsticks are needed for stage 15? (iv) Can 200 matchsticks form a stage exactly? Justify.

Expected answer / marking points: Theorem: Figure with arc PQ, centre O, point R on the remaining circle; join RO and extend to S [1]. In $\triangle OPR$, OP = OR $\Rightarrow \angle OPR = \angle ORP$; exterior $\angle POS = \angle OPR + \angle ORP = 2\angle ORP$ [1.5]. Similarly $\angle QOS = 2\angle ORQ$ [1]. Adding: $\angle POQ = 2(\angle ORP + \angle ORQ) = 2\angle PRQ$ [1.5].
OR (matchsticks): Table 6, 11, 16, 21 [1.5]; rule 5n + 1 [1.5]; stage 15: 76 [1]; $5n + 1 = 200 \Rightarrow n = 39.8$, not a whole number, so no [1].

Score: ___ / 40 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 23 October 2026 · 10 marks

Test ID: D2026-10-23

Science Ch 3 Plant tissues · Hindi वाक्य भेद · AI ICT Skills · Maths 5.8 Cyclic quadrilateral · Maths Ch 9 Propositions

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Q1 [1]

Question: Which plant tissue carries water from the roots to the leaves?

Expected answer / marking points: Xylem.

Q2 [3]

Question: अर्थ के आधार पर वाक्य-भेद लिखिए — (क) कृपया दरवाज़ा बंद कर दीजिए। (ख) भगवान तुम्हें सफल करे। (ग) वाह! कितना सुंदर दृश्य है!

Expected answer / marking points: (क) आज्ञावाचक (ख) इच्छावाचक (ग) विस्मयादिबोधक [1 × 3]

Q3 [1]

Question: Give one example each of an input device and an output device of a computer.

Expected answer / marking points: Input: keyboard/mouse/scanner/microphone; Output: monitor/printer/speaker [½ + ½].

Q4 [1]

Question: Which part of a computer is called its “brain”?

Expected answer / marking points: CPU (Central Processing Unit).

Q5 [2]

Question: PQRS is a cyclic quadrilateral with ∠P = 70° and ∠Q = 95°. Find ∠R and ∠S.

Expected answer / marking points: ∠R = 180° − 70° = 110°; ∠S = 180° − 95° = 85° [1 + 1].

Q6 [2]

Question: Write the converse of: “If a triangle is equilateral, then each of its angles is 60°.” Is the converse true?

Expected answer / marking points: Converse: If each angle of a triangle is 60°, then it is equilateral [1]. True [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 24 October 2026 · 10 marks

Test ID: D2026-10-24

SST Ch 2 Earth's surface · Maths Ch 2 Linear Polynomials · Hindi पाठ 4 (प्रश्न) · Maths 10.1 Averages

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Q1 [1]

Question: Which agent of gradation forms moraines?

Expected answer / marking points: Glaciers.

Q2 [2]

Question: How is a delta formed? Why are deltas good for farming?

Expected answer / marking points: River slows at its mouth and deposits sediment, forming fan-shaped land [1]; deposited silt makes it fertile [1].

Q3 [2]

Question: Riya has ₹500 and saves ₹150 every month. Write the amount after n months as a linear pattern and find the amount after 8 months.

Expected answer / marking points: A = 500 + 150n [1]; n = 8: ₹1,700 [1].

Q4 [2]

Question: लता जी के पिताजी बच्चों की शरारत पर कैसे पेश आते थे? इससे उनके स्वभाव के बारे में क्या पता चलता है?

Expected answer / marking points: कुछ कहते नहीं थे, बस गंभीरता से देखते थे, फिर पूछते "समझ गए न?" [1]। अनुशासन और स्नेह का संतुलन [1]।

Q5 [3]

Question: A student's three assessments have weights 3, 2 and 5. Her scores are 60%, 64% and 73%. Find her weighted mean score.

Expected answer / marking points: (60 × 3 + 64 × 2 + 73 × 5) ÷ 10 [1] = 673 ÷ 10 [1] = 67.3% [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Sunday, 25 October 2026 · 10 marks

Test ID: D2026-10-25

AI Data Acquisition · AI Data Exploration · SST Ch 3 Monsoon · Science 8.4–8.6 Atomic number

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Q1 [1]

Question: Name two sources from which data can be collected for an AI project.

Expected answer / marking points: Any two: surveys, interviews, observation, sensors, cameras, web scraping, APIs, open/government data portals.

Q2 [1]

Question: What does a system map show?

Expected answer / marking points: The elements of a system and the cause-and-effect relationships between them (shown with arrows).

Q3 [1]

Question: Why should data for an AI project come from reliable and fair sources?

Expected answer / marking points: Wrong or one-sided (biased) data makes the AI model give wrong or unfair results.

Q4 [2]

Question: Why do we explore data using graphs before building an AI model? Name two types of graphs.

Expected answer / marking points: To see patterns, trends and errors/missing values quickly and decide which data and model to use [1]. Bar graph, line graph, pie chart, histogram, scatter plot (any two) [1].

Q5 [1]

Question: In which months does the south-west monsoon bring rain to India?

Expected answer / marking points: June to September.

Q6 [1]

Question: Write one way in which your family can reduce its carbon footprint.

Expected answer / marking points: Save electricity, public transport/cycle, plant trees, renewable energy, less waste (any one).

Q7 [1]

Question: Write the symbol of sodium.

Expected answer / marking points: Na

Q8 [2]

Question: An atom has 17 protons and 18 neutrons. Find its atomic number and mass number.

Expected answer / marking points: Atomic number 17 [1]; mass number 35 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: